

Jason Tunstill

Senior Software Engineer | C++ | Python | Qt | Ruby on Rails | Full Stack | Linux | Windows

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SUMMARY

Senior full-stack software engineer with 11+ years designing and delivering production software across desktop, backend, and engineering automation systems. Expertise includes C++, Python, Qt, Linux, CI/CD, and API development, with a strong track record of modernizing legacy systems, architecting extensible software platforms, solving complex debugging challenges, and mentoring engineers.

TECHNICAL SKILLS

Languages: C++, C, Python, C#, Ruby, JavaScript, SQL, HTML/CSS, Bash, PowerShell, LabVIEW

Frameworks & Technologies: Qt, QML, PySide6, Ruby on Rails, Flask, FastAPI, SWIG, FLTK, jQuery, REST APIs, ZeroMQ

Desktop Application Development: Cross-Platform Development, GUI Design, Native Windows Applications, Linux Applications, Multithreaded Programming, Software Architecture

Build, DevOps & CI/CD: Git, GitHub, GitLab, GitLab CI, Azure DevOps, Azure Pipelines, Docker, CMake, Conan, QtIFW, Build Automation, Release Management

Databases: PostgreSQL, MySQL, MariaDB

Operating Systems: Linux (Ubuntu, Debian, CentOS, Oracle Linux, RHEL), Windows 10/11, Windows Server

Engineering & System Integration: API Design, System Integration, Hardware Interface Development, National Instruments DAQmx, DevExpress XtraGrid, FlexNet Publisher, License Management Systems, IPC (ZeroMQ), JSON, Client/Server Architecture

Software Engineering Practices: Object-Oriented Programming/Design (OOP/OOD), Design Patterns, Debugging, Performance Optimization, Code Reviews, Refactoring, Technical Documentation, Agile/Scrum, Mentoring, Cross-Functional Collaboration

AI-Assisted Development: GitHub Copilot, Claude, ChatGPT (GPT-5.x), AI-Assisted Debugging, Code Generation, Codebase Analysis

EXPERIENCE

Software Engineer — Amentum, Tullahoma, TN | March 2026 – Present

- Redesigned a C#/ .NET API integrating LabVIEW applications with DevExpress XtraGrid, resolving DLL loading and cross-threading issues while significantly improving application stability.
- Increased real-time display performance from approximately 0.7 updates/sec to 10 updates/sec (15× improvement) by redesigning data exchange between LabVIEW and DevExpress XtraGrid.
- Design and implement new features for LabVIEW-based mission-critical test and data acquisition applications, enhancing usability and expanding operational capabilities.
- Collaborate with cross-functional engineering teams to modernize legacy LabVIEW applications through improved software architecture, UI responsiveness, and maintainability.

Software Developer — Apex Turbine (formerly Experimental Design & Analysis Solutions), Spring Hill, TN | January 2015 – January 2026

- Developed new features and maintained 6+ cross-platform C++ engineering applications with Qt and FLTK for real-time data acquisition, analysis, and visualization on Windows and Linux.
- Developed C++ hardware integration plugins for National Instruments, VTI, Mecalc, and other DAQ systems, implementing channel configuration, timing, triggering, buffered acquisition, and error handling through vendor driver APIs.
- Developed a QML desktop client that communicated with a Python Flask server through REST endpoints for application services and data exchange.
- Architected a cross-platform Python licensing daemon integrating modernized C++17 FlexNet Publisher wrappers through SWIG and a custom ZeroMQ REQ/REP JSON protocol; provided a common licensing API, CLI, and PySide6 GUI used by 6 desktop applications.
- Implemented and maintained the response side of the OpenCode offline activation/deactivation workflow in Ruby on Rails, managing license counts and expirations while supporting FlexNet Trusted Storage without requiring direct Internet access on licensed systems.
- Served as sole developer and maintainer of a legacy Ruby on Rails business platform; upgraded Rails 4.1 through 6.0, migrated MariaDB to PostgreSQL, and moved deployment from a CentOS VM to Docker in Microsoft Azure.
- Built and enhanced internal timekeeping, CRM, contract management, inventory, quoting, and engineering workflows, including PDF/Excel quote generation, jQuery/AJAX interfaces, scheduled automation, and database/query optimization.
- Designed v1/v2 REST APIs and webhooks for bidirectional synchronization between customer support tickets in Rails and Azure DevOps work items, including ticket creation, status, priority, and comments.
- Created and maintained GitLab CI and Azure DevOps pipelines for Windows/Linux build, packaging, artifacts, installers, and releases; maintained QtIFW installers and troubleshoot build, release, and customer installation failures.
- Co-developed a Python/TypeScript build CLI integrating CMake configuration, compilation, Windows code signing, install staging, Python wheel generation, packaging, and Azure Pipelines artifact creation.
- Performed complex debugging across C++, Python, and Ruby on Rails codebases; conducted merge-request reviews, mentored junior developers and interns, and helped onboard engineers to mature codebases.
- Worked directly with dozens of customer teams on requirements, bug reproduction, crash/log analysis, hardware and data troubleshooting, hotfixes, software deployment, and technical support.

Associate Engineer / Project Lead — QualTech NP (Curtiss-Wright), Huntsville, AL | July 2009 – January 2015

- Led multiple electrical penetration assembly (EPA) projects from planning through completion, coordinating engineering activities, schedules, documentation, and customer deliverables.
- Developed assembly procedures, manufacturing documentation, and acceptance test procedures supporting production and qualification of electrical penetration assemblies.
- Designed qualification test procedures and technical reports for research and development programs in accordance with IEEE 317 and applicable industry standards.
- Developed qualification test procedures and analyzed test results for LOCA, seismic, insulation resistance, helium leak, dielectric, and environmental testing in accordance with IEEE 317 requirements.
- Tracked project progress, managed engineering documentation, and coordinated cross-functional teams to meet technical, quality, and schedule requirements.
- Prepared technical reports, analyzed test data, and documented qualification results supporting product certification and customer acceptance.

EDUCATION

Bachelor of Science, Computer Engineering — University of Alabama in Huntsville | May 2014